

KIS-Style Clinical Data Model: SQL to HL7 v2 to FHIR R4

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![DBeaver](https://img.shields.io/badge/DBeaver-26.2-382923?logo=dbbeaver&logoColor=white)
![HL7 v2](https://img.shields.io/badge/HL7-v2.5_ADT^A01-0072BC)
![FHIR](https://img.shields.io/badge/FHIR-R4-E8590C)
![LOINC](https://img.shields.io/badge/LOINC-8867--4-2F9E44)
![Data](https://img.shields.io/badge/Data-Synthetic_only-lightgrey)

A compact, end-to-end interoperability project. It models patient, admission and vital-sign data the way a hospital information system (KIS, e.g. ORBIS-type systems) organises it, and then traces that data into an HL7 v2 admission message and FHIR R4 resources.

> **Scope note:** This is a KIS-style data model built from my knowledge of how hospital systems like ORBIS structure data. It is not an ORBIS installation and uses no vendor software or real patient data.

1. Problem

Hospital information systems store clinical data in internal, vendor-specific tables. Surrounding systems (laboratory, radiology, ePA, MVZ software, research platforms) need the same data in standard formats.

- **Different models:** Every KIS has its own structure for patients, encounters and observations.
- **Two generations of standards:** Legacy interfaces use HL7 v2 (pipe-delimited messages). New requirements (ISiK, ePA, KBV profiles) use FHIR R4.
- **Silent mapping errors:** A ward in the wrong segment, a lost identifier, or a measurement without a code can go unnoticed until it affects a clinical decision or a billing process.

Question this project answers: How can one clinical record be followed from the database, through an HL7 v2 message, to FHIR R4, with every field traceable?

2. Approach

Step 1: Relational KIS-style data model (SQLite, DBeaver)

Three linked tables mirror the core of a hospital record:

Table	Purpose	Key fields
patient	Master data	patient_id (PK), last_name, first_name, birth_date, sex
encounter	Hospital stay	encounter_id (PK), patient_id (FK), ward, room, admit_datetime, status
observation	Clinical measurement	obs_id (PK), encounter_id (FK), loinc_code, display, value, unit, obs_datetime

Design decisions:

- 'NOT NULL' on keys and names, so incomplete records are rejected at the source.
- Foreign keys ('PRAGMA foreign_keys = ON') so no observation exists without an encounter and no encounter without a patient.
- Measurements carry a **LOINC code** ('8867-4', heart rate) so the value is machine-readable and comparable across systems.

A single JOIN returns the complete clinical picture of the patient:

```
```sql
SELECT p.patient_id, p.last_name, p.first_name, p.birth_date,
 e.encounter_id, e.ward, e.room,
 o.loinc_code, o.display, o.value, o.unit
FROM patient p
JOIN encounter e ON e.patient_id = p.patient_id
JOIN observation o ON o.encounter_id = e.encounter_id;
```
```

Step 2: HL7 v2.5 admission message (ADT^A01)

```
MSH|^~\&|ORBIS|KLINIKUM_DEMO|LAB|KLINIKUM_DEMO|20260930103000||ADT^A01^ADT_A01|MSG001|P|2.5
EVN|A01|20260930103000
PID|1||100001^^^KLINIKUM^MR||Mustermann^Max||19800512|M||Musterstrasse 1^^Berlin^^10115^DE
PV1|1|I|STA1^101^A||1234^Schmidt^Anna||MED|||||V0001
^^^
```

| Segment | Content |
|---------|---|
| MSH | Sender, receiver, timestamp, message type, HL7 version |
| EVN | Trigger event (A01 = patient admission) |
| PID | Patient identity: MRN, name, birth date, sex, address |
| PV1 | Visit: inpatient class, ward/room/bed (PV1-3), attending physician (PV1-7), service (PV1-10), visit number (PV1-19) |

Step 3: Field-level mapping, KIS to HL7 v2 to FHIR R4

| KIS database field | HL7 v2 | FHIR R4 |
|--------------------------------|-----------------|---------------------------|
| patient.patient_id | PID-3 | Patient.identifier |
| patient.last_name / first_name | PID-5 | Patient.name |
| patient.birth_date | PID-7 | Patient.birthDate |
| patient.sex | PID-8 | Patient.gender |
| encounter.ward / room | PV1-3 | Encounter.location |
| encounter.encounter_id | PV1-19 | Encounter.identifier |
| encounter.admit_datetime | EVN-2 / PV1-44 | Encounter.period.start |
| observation.loinc_code | OBX-3 (ORU^R01) | Observation.code |
| observation.value / unit | OBX-5 / OBX-6 | Observation.valueQuantity |

Step 4: FHIR R4 resources (Postman, public HAPI test server)

'Patient', 'Encounter' and 'Observation' JSON resources are created with 'POST' requests and linked by reference. A 'GET Patient/{id}/\$everything' call returns them as one Bundle.

3. Challenges and Edge Cases

| Challenge | Why it matters | How it is handled |
|-------------------------|---|-------------------|
| HL7 v2 field positions | A value one pipe off lands in the wrong field (e.g. visit number in 'PV1-16' instead of 'PV1-19'). Receiving systems then read wrong data without any error. Fields counted explicitly. Attending doctor in 'PV1-7', service in 'PV1-10', visit number in 'PV1-19'. | |
| Identifier handling | The KIS patient ID, the encounter number and the FHIR resource ID are three different things. MRN in 'PID-3' and 'Patient.identifier'; visit number in 'Encounter.identifier'; FHIR server IDs are generated separately. | |
| Date formats | KIS stores '2026-09-30 10:30', HL7 v2 needs '20260930103000', FHIR needs ISO 8601 with time zone. Conversion rule documented per field. | |
| Referential integrity | Orphan records break downstream joins and exports. Foreign keys enforced in the schema. | |
| Coding of measurements | Free-text names ("Puls", "HR", "Herzfrequenz") cannot be compared. LOINC code stored with every observation. | |
| One source, two targets | HL7 v2 sends events (ADT, ORU); FHIR stores resources. The same fact appears in two shapes. Mapping table keeps both representations traceable. | |
| Privacy | Patient data falls under DSGVO Art. 9. Synthetic data only; public test server never receives real data. | |

4. Plan to Extend

- ORU^R01 message for the heart-rate observation (OBX segment).
- Python transformation script that reads the SQLite tables and generates the HL7 v2 message and FHIR JSON automatically.
- Validation of the FHIR resources against ISiK / KBV profiles on Simplifier.net.

4. ****Integration engine test**** with Mirth Connect (receive ADT^A01, transform to FHIR).
5. ****Additional edge cases:**** multiple encounters per patient, transfer (ADT^A02) and discharge (ADT^A03), missing or invalid codes.

5. Why This Matters for Healthcare

- ****Patient safety:**** Correct identifiers, locations and coded values make sure results and documents reach the right patient.
- ****Interoperability:**** Lab, radiology, MVZ and ePA systems exchange data without manual re-entry.
- ****Migration readiness:**** Hospitals are moving from HL7 v2 interfaces to FHIR R4 (ISiK, ePA). A documented field mapping is the starting point for every such migration.
- ****Data quality at the source:**** Constraints and coded values stop errors before they reach downstream systems.
- ****Compliance:**** Structured, traceable data flows support DSGVO Art. 9 and gemSpec_ePA requirements.
- ****Analytics and research:**** LOINC-coded observations can be reused for reporting, quality assurance and AI projects.

6. Tools and Standards

| Area | Tools / Standards |
|---|---|
| Database | SQLite, DBeaver 26.2 |
| Messaging | HL7 v2.5 (ADT^A01), Notepad++ |
| Modern standard | FHIR R4, HAPI FHIR public test support, Postman |
| Terminology | LOINC (`8867-4` Heart rate) |
| Concepts | Relational modelling, referential integrity, field mapping, message structure, terminology binding |

7. Repository Contents

- └─ sql/ kis_schema_and_data.sql
- └─ hl7/ admission_A01.hl7
- └─ fhir/ patient.json, encounter.json, observation.json
- └─ screenshots/
- └─ README.md

8. Data Privacy

All data is ****fictional and synthetic****. No real patient data is used.